



# Gen AI and Agentic AI in Production

Week 1  
Day 2

# 4 Week Journey to Prod-Grade

## Week 4: Agentic AI in Production

Multi-Agent  
Architecture

Capstone:  
Agents on  
AWS

Capstone:  
Frontend

Observability  
Monitoring  
Security

Agentic AI  
Platforms

## Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on  
Azure with  
MCP

Cyber AI on  
GCP with  
MCP

AWS  
SageMaker

Data Ingest  
S3 Vectors

Data Pipes  
with MCP

## Week 2: AI Platform Engineering on AWS

AWS  
Architecture

The Digital  
Twin Mk 2

AWS  
Bedrock

Terraform

Github  
Actions

## Week 1: SaaS Generative AI in Production

First Gen AI  
Deployment

Frontend &  
Backend

Auth &  
Subscription

Healthcare  
SaaS App

Deploy to  
AWS

# Barely touch the surface



Some of you may be frontend engineers; some new to it

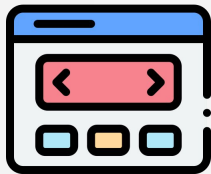
I will cover the basics to give you intuition

Review the code and ask an LLM to explain it

# Preliminaries

## Frontend vs Backend in a web app

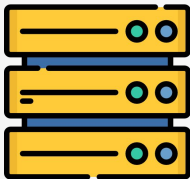
### Frontend



#### Runs in user's browser

HTML (structure)  
CSS (appearance)  
Javascript (interactivity)

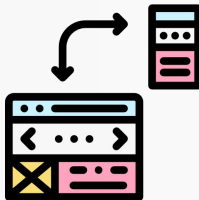
### Backend



#### Runs on the server

Business logic, database  
access, LLM calling, other  
APIs, secrets

### Frontend+Backend



#### API contract over HTTP

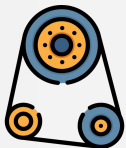
The backend exposes HTTP  
endpoints, called by the frontend;  
the backend can also serve the  
frontend code to browser

## So wait - how does Gradio fit in?

*Gradio is backend framework that automatically generates & serves a frontend*

With apologies to the Frontend developers amongst us..

# Frontend progression



## Vanilla HTML

HTML + CSS + Javascript + low level libraries like jQuery  
The “DOM” = object representation of the page



## Frontend Frameworks like React

React, Vue, Angular, Svelte - reusable components  
SPA - loads page once and dynamically updates  
Come in Javascript variants, and its sister language Typescript



## Application Frameworks like Next.js

Adds routing, data fetching, server-side rendering and more  
Tooling, such as bundling  
JS or TS

# React - JS / TS frontend framework

## Component-based



### Reusable components

Building blocks with markup, styles logic

## Declarative



### Describe UI given state

React efficiently updates the DOM with any changes

## Props & State



### Get props, manage state

When either change, React re-renders the component and components it contains

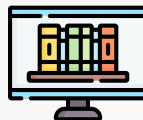
## JSX / TSX



### Convenient format

Blends HTML and JS/TS in a simple, easy-to-read way to increase productivity

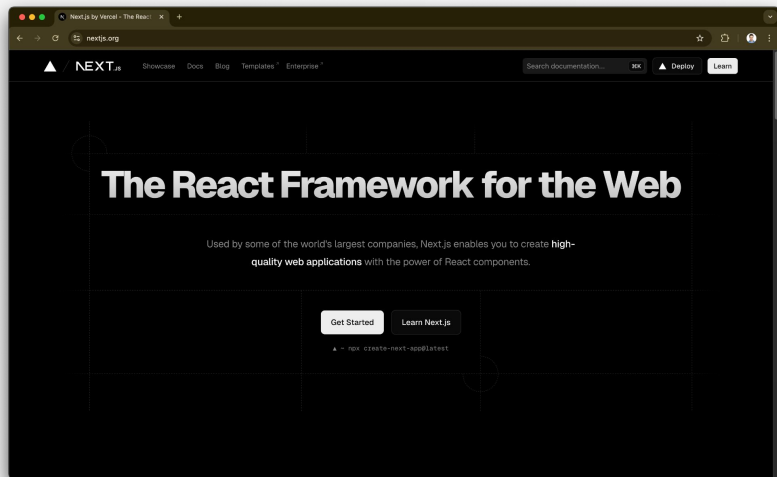
## Ecosystem



### Incredibly Rich

Forms, Routing, UI component libraries like Material UI, Chakra UI

# NextJS from Vercel



**Substantial functionality out of the box**

**Includes 2 approaches for Routing requests for pages:**

**Pages router** (pages/) is simpler, trusted, very common

**App Router** (app/) is newer, more powerful

**Pages can be rendered client-side or server-side**

**Includes built-in tooling to transpile & bundle**

And just to complete the story

# Python Backend Frameworks



## Django

Heavyweight, “batteries included” framework  
ORM, authentication, templating, admin interface



## Flask

Micro-framework - routing and request handling  
Lightweight and quick to start, but requires add-ons



## FastAPI

Modern, async, built on Starlette and Pydantic  
Optimized for APIs

# Course repo - a bit confusing



**Start with the main repo: production**

git clone <https://github.com/ed-donner/production.git>

**This has a week1 folder with instructions for each day**

We will create new repos from scratch like **saas** (today)

**It has a guides folder with self-study guides**

**And it has community\_contributions folder**

**In weeks 3 and 4, we will move to other course repos**

TO THE LAB!



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10%  
Complete